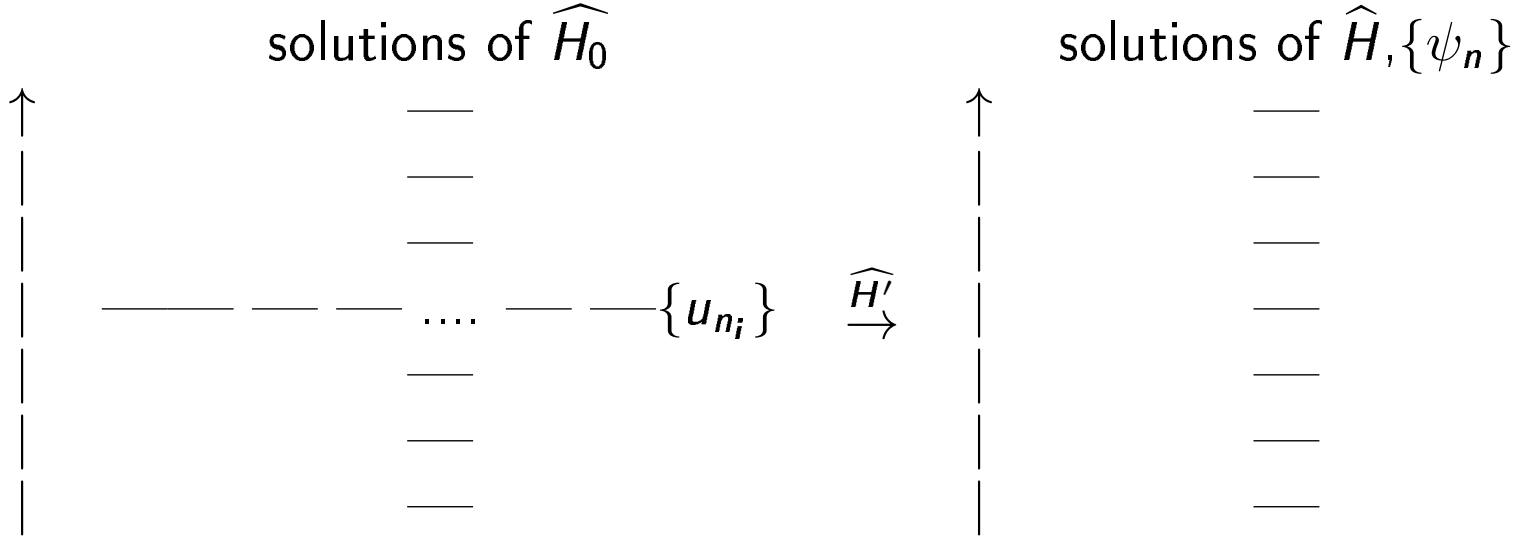
Degenerate perturbation theory

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For situations when there are $d > 1$ solutions of $\widehat{H_0}$, corresponding to a unique energy E_n^0, then we use the method of degenerate perturbation theory.



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If we started with these proper linear combinations, then we would have obtained a diagonal H'_{ij} matrix.

For perturbation corrections, we start again but instead of using the original u_{n_i}s, we use the χ_{n_i}s.

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(i) Find the unperturbed energy levels, noting the degeneracy.
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$u_0^I = \frac{1}{\sqrt{2}}(\uparrow\uparrow - \uparrow\downarrow) \equiv \frac{1}{\sqrt{2}}(\lvert 00\rangle - \lvert 01\rangle)$		
$u_0^{II} = \frac{1}{\sqrt{2}}(\downarrow\uparrow + \downarrow\downarrow) \equiv \frac{1}{\sqrt{2}}(\lvert 10\rangle + \lvert 11\rangle)$		
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Example : A system of two spins is described by the Hamiltonian $H = H_0 + H'$

$$H_0 = \frac{A}{2}\{\sigma_z(1) + \sigma_x(2)\}; \quad H' = B\{\sigma_x(1)\sigma_z(2) + \sigma_z(1)\sigma_x(2)\}$$

(i) Find the unperturbed energy levels, noting the degeneracy.
(ii) compute the energy shifts to first order in B.

Ans. $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$; $\hat{\sigma}_z \begin{matrix} \uparrow \\ \downarrow \end{matrix} = \begin{matrix} \uparrow \\ -\downarrow \end{matrix}$;and $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$; $\hat{\sigma}_x \begin{matrix} \uparrow \\ \downarrow \end{matrix} = \begin{matrix} \downarrow \\ \uparrow \end{matrix}$;

Eigenstates and eigenvalues of σ_z		Eigenstates and eigenvalues of σ_x	
$\uparrow \equiv \lvert 0\rangle$	1	$\frac{1}{\sqrt{2}}(\uparrow + \downarrow) \equiv \frac{1}{\sqrt{2}}(\lvert 0\rangle + \lvert 1\rangle)$	1
$\downarrow \equiv \lvert 1\rangle$	-1	$\frac{1}{\sqrt{2}}(\uparrow - \downarrow) \equiv \frac{1}{\sqrt{2}}(\lvert 0\rangle - \lvert 1\rangle)$	-1

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$u_0^I = \frac{1}{\sqrt{2}}(\uparrow\uparrow - \uparrow\downarrow) \equiv \frac{1}{\sqrt{2}}(\lvert 00\rangle - \lvert 01\rangle)$	$E_0^{(0)} = \frac{A}{2}(1-1) = 0$	—— 0
$u_0^{II} = \frac{1}{\sqrt{2}}(\downarrow\uparrow + \downarrow\downarrow) \equiv \frac{1}{\sqrt{2}}(\lvert 10\rangle + \lvert 11\rangle)$	$E_0^{(0)} = \frac{A}{2}(-1+1) = 0$	—— $-A$
$u_{-2} = \frac{1}{\sqrt{2}}(\downarrow\uparrow - \downarrow\downarrow) \equiv \frac{1}{\sqrt{2}}(\lvert 10\rangle - \lvert 11\rangle)$	$E_{-2}^{(0)} = \frac{A}{2}(-1-1) = -A$	

(ii) For the nondegenerate levels $E_2^{(0)}$, $E_{-2}^{(0)}$, the first order shifts are:

$$E_2^{(1)} = \langle u_2|H'|u_2\rangle = \frac{B}{2}(\langle 00| + \langle 01|)\sigma_x(1)\sigma_z(2) + \sigma_z(1)\sigma_x(2)(|00\rangle + |01\rangle)$$

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For the unperturbed levels at $E_0^{(0)} = 0$, we use degenerate methods. First, get the matrix H′:

$$\langle u_0^{II}|H'|u_0^{II}\rangle = \frac{B}{2}(\langle 10| + \langle 11|)\sigma_x(1)\sigma_z(2) + \sigma_z(1)\sigma_x(2)(|10\rangle + |11\rangle)$$

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The shifts are obtained from the eigenvalues of the matrix: $\begin{pmatrix} -B & B \\ B & -B \end{pmatrix}$

i. e., 0 and $-2B$

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what are the new wavefunctions?